Quantum Query Lower Bounds

Hybrid, Polynomial, and Adversary Methods

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Computational complexity (CC)

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- Computational models (Church-Turing Thesis, for polynomial time)

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Generally speaking, very hard.

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Generally speaking, very hard.

- Sublinear: information-theoretic argument and unconditional (This lecture)
- Superlinear: conjectures (assume $P \neq NP$), conditional (See next lecture)

Plan for the lecture

1. Quantum Query Model
2. Hybrid Method
3. Polynomial Method
4. Adversary Method
5. Frontier: Recording / Compressed Oracle
6. Wrap-up

Goal. Three core techniques to prove *nobody* can solve a problem in few quantum queries:

- Hybrid — the original (BBBV '97); gives $\sqrt{N}$ for search.
- Polynomial — algebra: Pr[accept] is a polynomial in x.
- Adversary — spectral; tight for most problems we care about.

Plus a brief look at the *recording method* (Zhandry '18) as a frontier topic.

Quantum query model

Input. $x = x_1 x_2 \cdots x_N \in \{0,1\}^N$, accessed only through the oracle

$$O_x : \ |i\rangle\,|b\rangle \mapsto |i\rangle\,|b \oplus x_i\rangle\,, \qquad i \in [N],\ b \in \{0,1\}.$$

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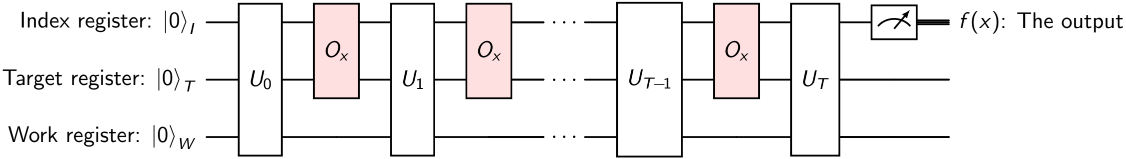
Algorithm. T input-independent unitaries $U_0, \ldots, U_T$ interleaved with queries:

$$|\psi_T^x\rangle \;=\; U_T\, O_x\, U_{T-1}\, O_x \cdots U_1\, O_x\, U_0\,|0\rangle\,.$$



Output: measurement on a designated register.

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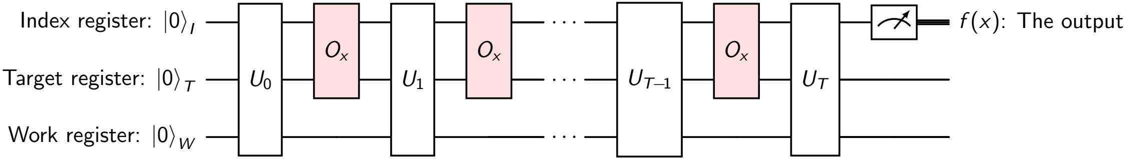

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Bounded-error query complexity.

$$Q_\epsilon(f) \;=\; \min\{\, T : \exists A \text{ s.t. } \Pr[A(x) = f(x)] \geq 1 - \epsilon \ \ \forall x \in \mathrm{dom}(f)\}.$$

We write $Q(f) := Q_{1/3}(f)$.

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  - Grover: $Q(\mathrm{OR}_N) = \Theta(\sqrt{N})$.
  - Simon: $\Theta(n)$ vs. classical $\Omega(2^{n/2})$.
  - Shor (period finding): quantum query polynomial; classical exponential.
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Cons.

- **Sometimes tight, sometimes not.** E.g. k-sum, collision-finding in structured inputs.
- **At most linear**: After reading all the input bits, the algorithm is supposed to obtain the result. (Superlinear? See next lecture)

Two useful equivalent oracles

We often use a phase oracle instead of the XOR oracle:

$$O_x^{\pm} |i\rangle = (-1)^{x_i} |i\rangle .$$

Equivalent up to one ancilla in state $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$:

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Why bother.

- Some arguments are cleaner with phase queries (polynomial method).
- some with XOR (adversary method).
- The conversion costs $O(1)$ queries.

Hybrid method — the idea (BBBV '97)

Bennett, Bernstein, Brassard and Vazirani: *Strengths and weaknesses of quantum computation*.

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Fix algorithm A with T queries. Let

$$|\phi_t^x\rangle = U_t \, O_x \, U_{t-1} \, O_x \cdots U_0 \, |0\rangle$$

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Query magnitude. The state *just before* the $(t+1)$st oracle call is $|\phi_t^x\rangle$. Its weight on index i is

$$q_i(t,x) \;=\; \left\|(|i\rangle\langle i| \otimes \mathrm{Id})\, |\phi_t^x\rangle\right\|^2 .$$

One-query perturbation lemma

Lemma. Let $|\phi\rangle$ be the state just before a query. If x, y differ on $S \subseteq [N]$, then

$$\left\| O_x |\phi\rangle - O_y |\phi\rangle \right\|^2 \le 4 \sum_{i \in S} q_i(t, x).$$

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Proof. Write $|\phi\rangle = \sum_{i,b,w} \alpha_{i,b,w} |i, b, w\rangle$. Then

$$O_x |\phi\rangle - O_y |\phi\rangle \ = \sum_{i \in S,\, b,\, w} \alpha_{i,b,w} \big[|i, b \oplus x_i, w\rangle - |i, b \oplus y_i, w\rangle \big].$$

Each bracket has norm ≤ 2 (since $x_i \neq y_i$). The vectors are orthogonal across different (i, b, w), so

$$\left\| O_x |\phi\rangle - O_y |\phi\rangle \right\|^2 \ \leq \ 4 \sum_{i \in S,\, b,\, w} |\alpha_{i,b,w}|^2 \ = \ 4 \sum_{i \in S} q_i(t, x). \quad \square$$

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Lemma (BBBV). For inputs x, y differing on $S \subseteq [N]$,

$$\| |\phi_T^x\rangle - |\phi_T^y\rangle \| \ \leq \ 2 \sum_{t=0}^{T-1} \sqrt{\sum_{i \in S} q_i(t,x)}.$$

Equivalently (Cauchy–Schwarz), $\| |\phi_T^x\rangle - |\phi_T^y\rangle \|^2 \leq 4T \sum_{t=0}^{T-1} \sum_{i \in S} q_i(t,x)$.

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Proof. Define the *hybrid state* $|H_k\rangle$: queries $1, \ldots, k$ use O_y; queries $k+1, \ldots, T$ use O_x:

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Triangle inequality. $\| |\phi_T^x\rangle - |\phi_T^y\rangle \| \leq \sum_{k=0}^{T-1} \| |H_k\rangle - |H_{k+1}\rangle \|\,.$

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$|H_k\rangle$ and $|H_{k+1}\rangle$ differ only at query $k+1$: $(O_x - O_y)$ acts on $|\phi_k^x\rangle$ (unitaries preserve norms). By the perturbation lemma: $\| |H_k\rangle - |H_{k+1}\rangle \| \le 2\sqrt{\sum_{i \in S} q_i(k,x)}$. □

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Step 1: distinguishing $\Rightarrow$ large state difference. If two states $|\alpha\rangle, |\beta\rangle$ yield acceptance probabilities differing by δ under the *same* measurement, then $\| |\alpha\rangle - |\beta\rangle \| \geq \delta$.

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Hence

$$\left\| |\phi_T^{e_k}\rangle - |\phi_T^{z}\rangle \right\|^2 \;\geq\; 1/9.$$

Application: OR lower bound via hybrid (cont.)

Step 2: average over k.

$$\sum_{k=1}^{N} \left\| |\phi_T^{e_k}\rangle - |\phi_T^{z}\rangle \right\|^2 \;\le\; 4T \sum_{k=1}^{N} \sum_{t} q_k(t,z) \;=\; 4T \sum_{t} \underbrace{\sum_{k=1}^{N} q_k(t,z)}_{=1} \;=\; 4T^2.$$

(Using BBBV with $S = \{k\}$ for each k, and $\sum_k q_k(t,z) = 1$.)

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Step 3: combine.

$$N \cdot \frac{1}{9} \leq \sum_{k=1}^{N} \left\| |\phi_T^{e_k}\rangle - |\phi_T^{z}\rangle \right\|^2 \leq 4T^2.$$

Hence $T \geq \sqrt{N}/6 = \Omega(\sqrt{N})$. □

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Remarks.

- The bound is tight: Grover achieves $O(\sqrt{N})$.
- The constants can be sharpened (Zalka '99) to give the exactly tight $\pi\sqrt{N}/4$.

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$\Rightarrow$ Motivates more powerful methods.

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Theorem. If A is a quantum algorithm with T queries, then

$$P(x) := \Pr[A \text{ accepts } x]$$

is a multilinear polynomial in $x_1, \ldots, x_N \in \{0,1\}$ of degree at most $2T$.

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Corollary.

$$Q_\epsilon(f) \;\geq\; \tfrac{1}{2}\,\widetilde{\deg}_\epsilon(f),$$

where $\widetilde{\deg}_\epsilon(f)$ is the minimum degree of a polynomial ϵ-approximating f pointwise on $\mathrm{dom}(f) \subseteq \{0,1\}^N$.

Proof of the BBCMdW theorem (I)

Lemma. For every basis state $|j\rangle$ of the workspace, the amplitude

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Proof by induction on t.

Base case ($t = 0$). $|\phi_0^x\rangle = U_0\,|0\rangle$, independent of x. Each amplitude is a constant (degree 0). ✓

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Conclusion. The acceptance probability is

$$P(x) = \sum_{j \in \text{accept}} |\alpha_j^x|^2 \implies \deg P \leq 2T.$$

Multilinearity: $x_i^2 = x_i$ on $\{0,1\}$, so $|\alpha|^2$ remains multilinear. $\square$

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Markov's inequality. If $\tilde{P}$ is a degree-d polynomial with $|\tilde{P}(x)| \le M$ on $[a, b]$, then

$$\max_{x \in [a,b]} |\tilde{P}'(x)| \;\le\; \frac{2d^2 M}{b - a}.$$

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Let $\tilde{P}$ be a degree-d univariate polynomial $1/3$-approximating g on $\{0, \ldots, N\}$, so:

$$\tilde{P}(0) \leq 1/3, \qquad \tilde{P}(k) \geq 2/3 \text{ for } 1 \leq k \leq N, \qquad |\tilde{P}(k)| \leq 4/3 \text{ throughout.}$$

On the interval $[0, 1]$, $\tilde{P}$ jumps from $\leq 1/3$ at 0 to $\geq 2/3$ at 1, so $\max_{[0,N]} |\tilde{P}'| \geq 1/3$.

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$f = \mathrm{OR}_N$: $g(0) = 0$, $g(k) = 1$ for $k \geq 1$.

Let $\tilde{P}$ be a degree-d univariate polynomial 1/3-approximating g on $\{0, \ldots, N\}$, so:

$$\tilde{P}(0) \leq 1/3, \qquad \tilde{P}(k) \geq 2/3 \text{ for } 1 \leq k \leq N, \qquad |\tilde{P}(k)| \leq 4/3 \text{ throughout.}$$

On the interval $[0,1]$, $\tilde{P}$ jumps from $\leq 1/3$ at 0 to $\geq 2/3$ at 1, so $\max_{[0,N]} |\tilde{P}'| \geq 1/3$.

Markov's inequality on $[0, N]$**:**

$$\tfrac{1}{3} \leq \max_{[0,N]} |\tilde{P}'| \leq \frac{2d^2 \cdot (4/3)}{N} \implies d \geq \sqrt{N/8}.$$

$$\boxed{\widetilde{\deg}(\mathrm{OR}_N) = \Omega(\sqrt{N})} \quad \Rightarrow \quad Q(\mathrm{OR}_N) = \Omega(\sqrt{N}).$$

Match with Nisan–Szegedy: $\widetilde{\deg}(\mathrm{OR}_N) = \Theta(\sqrt{N})$.

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Progress measure. Track how distinguishable the algorithm's states are across all paired inputs:

$$W_t \;=\; \sum_{(x,y)\in R} \langle \phi_t^x | \phi_t^y \rangle\,.$$

When W_t is large, the algorithm has not yet learned enough to tell x from y. Each query to O can only decrease W_t by a bounded amount — so many queries are needed.

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Goal. Show each query changes W_t by a bounded amount $\Rightarrow$ many queries needed.

Bounding $|W_{t+1} - W_t|$ (the key step)

Define for each $i \in [N]$, $x \in X$, $y \in Y$:

$$\ell_{x,i} = |\{y : (x,y) \in R,\ x_i \neq y_i\}|, \qquad \ell'_{y,i} = |\{x : (x,y) \in R,\ x_i \neq y_i\}|.$$

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Step 1. Only indices where $x_i \neq y_i$ contribute (same idea as the perturbation lemma):

$$|W_{t+1} - W_t| \ \leq \sum_{(x,y)\in R} \sum_{i:\, x_i \neq y_i} \| |\phi_t^x\rangle\, |_i \| \cdot \| |\phi_t^y\rangle\, |_i \| \,,$$

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Step 2. For fixed i, let $R_i = \{(x,y) \in R : x_i \neq y_i\}$. Note two counting facts:

$$\sum_{(x,y)\in R_i} q_i(t,x) \ = \ \sum_x q_i(t,x) \cdot \ell_{x,i} \ \leq \ \ell \sum_x q_i(t,x),$$

because each x appears in $\ell_{x,i} \leq \ell$ pairs. Similarly $\sum_{(x,y)\in R_i} q_i(t,y) \leq \ell' \sum_y q_i(t,y)$.

Bounding $|W_{t+1} - W_t|$ (cont.)

Step 3 (Cauchy–Schwarz on (x, y)). Apply C-S directly to the sum over pairs $(x, y) \in R_i$:

$$\sum_{(x,y)\in R_i} \sqrt{q_i(t,x)}\,\sqrt{q_i(t,y)} \le \sqrt{\sum_{(x,y)\in R_i} q_i(t,x)} \cdot \sqrt{\sum_{(x,y)\in R_i} q_i(t,y)}.$$

Substituting the bounds from Step 2:

$$\le \sqrt{\ell \textstyle\sum_x q_i(t,x)} \cdot \sqrt{\ell' \textstyle\sum_y q_i(t,y)} = \sqrt{\ell\,\ell'} \cdot \sqrt{\textstyle\sum_x q_i(t,x)} \cdot \sqrt{\textstyle\sum_y q_i(t,y)}.$$

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Step 4 (Cauchy–Schwarz on i). Sum over i and apply C-S:

$$|W_{t+1} - W_t| \;\le\; \sqrt{\ell\,\ell'} \sum_i \sqrt{\sum_x q_i(t,x)}\,\sqrt{\sum_y q_i(t,y)} \;\le\; \sqrt{\ell\,\ell'} \cdot \sqrt{|X|} \cdot \sqrt{|Y|},$$

using $\sum_i q_i(t,x) = 1$ for every x, so $\sum_i \sum_x q_i(t,x) = |X|$ (similarly $|Y|$).

Ambainis's adversary bound

After T queries:

$$\Omega(|R|) = |W_0 - W_T| \leq T \cdot \sqrt{\ell \, \ell'} \cdot \sqrt{|X| \cdot |Y|}.$$

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Define $m = \min_x |\{y : (x,y) \in R\}|$ and $m' = \min_y |\{x : (x,y) \in R\}|$ (min row/column degrees of R).
Then $|R| \geq m|X|$ and $|R| \geq m'|Y|$, so $|X| \leq |R|/m$, $|Y| \leq |R|/m'$:

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Theorem (Ambainis).

$$T \;=\; \Omega\!\left(\sqrt{\frac{m\,m'}{\ell\,\ell'}}\right).$$

Application: OR lower bound via adversary

Setup. $X = \{0^N\}, \quad Y = \{e_1, \ldots, e_N\}, \quad R = X \times Y.$

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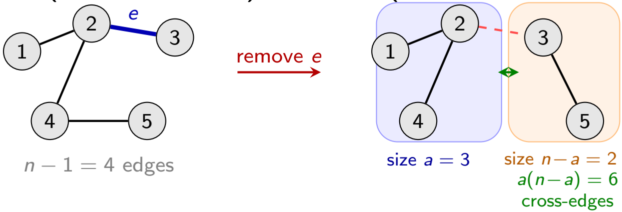
Apply the bound.

$$Q(\mathrm{OR}_N) \;=\; \Omega\left(\sqrt{\frac{m \cdot m'}{\ell \cdot \ell'}}\right) \;=\; \Omega\left(\sqrt{\frac{N \cdot 1}{1 \cdot 1}}\right) \;=\; \Omega(\sqrt{N}). \quad \square$$

Application: Graph connectivity

Problem. Given the adjacency matrix of an n-vertex graph ($N = \binom{n}{2}$ bits), is it connected?

$y \in Y$ **(tree, connected)** $\quad$ $x \in X$ **(forest, disconnected)**



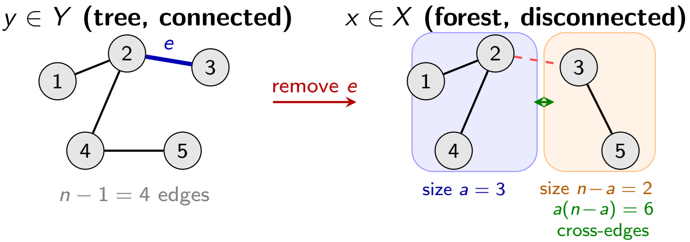
Adversary relation R:

$(x, y) \in R$ iff $y = x \cup \{e\}$
(differ on *exactly one* edge)

Key insight:

Each pair disagrees on 1 bit $\Rightarrow \ell = \ell' = 1$
But many pairs per input $\Rightarrow$ large m, m'

Application: Graph connectivity — computing parameters



Compute parameters.

- $m' = n - 1$: each tree y has $n - 1$ edges to remove, giving $n - 1$ paired forests.

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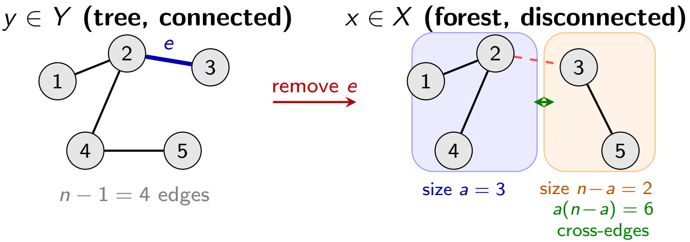

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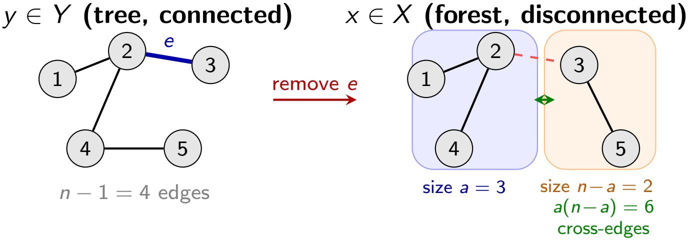

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- $\ell = \ell' = 1$: each pair $(x, y) \in R$ disagrees on exactly one bit (the added/removed edge).

Application: Graph connectivity — apply the bound

Apply the bound.

$$Q(\mathrm{CONN}_n) \;=\; \Omega\left(\sqrt{\frac{m \cdot m'}{\ell \cdot \ell'}}\right) = \Omega\left(\sqrt{\frac{(n-1)^2}{1}}\right) = \Omega(n).$$

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Remark. The tight bound is $\Theta(n^{3/2})$ (Dürr–Heiligman–Høyer–Mhalla '04)

General (negative-weight) adversary

Høyer–Lee–Špalek '07: allow Γ to have negative entries.

Definition. Let Γ be a $|\mathrm{dom}(f)| \times |\mathrm{dom}(f)|$ Hermitian matrix with $\Gamma[x, y] = 0$ whenever $f(x) = f(y)$. Let D_i be the 0/1 matrix with $D_i[x, y] = \mathbf{1}[x_i \neq y_i]$.

$$\mathrm{Adv}^{\pm}(f) = \max_{\Gamma \neq 0} \frac{\|\Gamma\|}{\max_i \|\Gamma \circ D_i\|},$$

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$\Rightarrow$ The negative-weight adversary *exactly characterises* bounded-error quantum query complexity.

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Frontier: the recording method (Zhandry '18)

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Comparison

Method	Best for	Tight?	Style
Hybrid	Search / unstructured oracles	Often (search)	Direct, geometric
Polynomial	Symmetric functions	Up to a polynomial gap	Algebraic
Adversary	Boolean functions, general	$\Theta(\mathrm{Adv}^{\pm})$ (neg. weights)	Spectral / SDP
Recording	QROM / Crypto	Many cases	Combinatorial

Take-aways.

- No single method dominates — the right tool depends on the problem.
- Adversary is tight in principle; polynomial and hybrid are often easier in practice.
- For cryptographic settings (random oracles), the recording method is the modern tool.

Further reading

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Questions?